



Development of a low-field benchtop NMR for ^{13}C polarimetry

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Abstract

Hyperpolarised ^{13}C magnetic resonance imaging (MRI) uses ^{13}C -labelled tracers for disease diagnosis and monitoring. Due to the low natural abundance of ^{13}C , hyperpolarisation is used to achieve detectable signal intensity. Polarimetry is required to evaluate and optimise tracer polarisation. Knowing the polarisation of the sample is also key during multi-centre studies for standardisation and quality control to correctly interpret metabolic information and evaluate image quality.

Some clinical hyperpolarisers, such as the GE SPINlab, which is currently used at the Oxford Centre for Clinical Magnetic Resonance Research (OCMR), lack the capability of liquid-state polarimetry and can only perform relative solid-state polarimetry of ^{13}C . Currently an MRI scanner is used for absolute liquid-state ^{13}C polarimetry, however this is inefficient and cannot be used for quality control during clinical scans. Commercially available benchtop devices for ^1H spectroscopy are also capable of ^{13}C polarimetry, are costly and over-optimised to solve this problem.

Building on a modular, open-source, low-field nuclear magnetic resonance (NMR) platform (NMRduino), we worked towards developing a benchtop NMR that could be used for absolute liquid-state ^{13}C polarimetry. A Helmholtz coil pair was built to generate a main magnetic field of 1.32 mT, corresponding to a resonance frequency of 14.1 kHz for ^{13}C . A solenoid was constructed for a transmit and receive coil with a loaded quality factor of 2.04 and a field-per-current of 4.05 mT/A_{peak}. Coil designs were simulated with Ansys Maxwell and High-Frequency Structure Simulator (HFSS) and compared to the experimental data. 3D modelling and printing were used for the circuit board case and the sample holder, including the coil winding form.

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Chapter 1

Introduction

1.1 Basics of magnetic resonance

Magnetic resonance imaging (MRI) scanners and nuclear magnetic resonance (NMR) spectrometers use multiple magnetic fields to gain information on the nuclear characteristics of the sample. A main magnetic field (B_0) determines the energy gap (ΔE) and resonance frequency (f_0) of nuclei with non-zero nuclear spin. When NMR active nuclei with a spin of $\frac{1}{2}$ are in an external magnetic field, their magnetic moments precess around B_0 , either parallel or antiparallel, resulting in $N_{\uparrow}$ nuclei in the lower-energy state and $N_{\downarrow}$ nuclei in the higher-energy state [1]. The polarisation of the sample (P_{sample}) is defined by the relative populations of these energy states according to Equation (1.1) [2]. The gyromagnetic ratio (γ) of NMR active isotopes is one of the factors governing the polarisation of the sample according to the Boltzmann distribution (Equation (1.2)), where h is the Planck constant, k is the Boltzmann constant, and T is the temperature. ^1H is the most frequently targeted isotope in magnetic resonance due to its high γ , high relative sensitivity and abundance (Table 1.1).

Table 1.1: Examples of nuclei used in MRI and NMR studies. Adapted from MRIQuestions.com [3], courtesy of Allen D. Elster and from Park et al. [4] under a CC BY 3.0 licence.

Nucleus	Spin	$\gamma/2\pi$ (MHz/T)	Relative Sensitivity	Abundance (%)
^1H	1/2	42.58	1.000	99.99
^{13}C	1/2	10.71	0.016	1.108
^{23}Na	3/2	11.26	0.083	100.0
^{31}P	1/2	17.24	0.066	100.0
^{129}Xe	1/2	11.78	0.021	26.44
^{15}N	1/2	-4.32	0.001	0.37

$$P_{\text{sample}} = \frac{N_{\uparrow} - N_{\downarrow}}{N_{\uparrow} + N_{\downarrow}} \quad (1.1)$$

$$\frac{N_{\downarrow}}{N_{\uparrow}} = e^{-\frac{\Delta E}{kT}} = e^{-\frac{\gamma \hbar B_0}{2\pi kT}} \quad (1.2)$$

Shimming fields are used to correct for the inhomogeneities in B_0 . Shielding minimises the signal interference with other nearby electronic components. Gradient coils are used for the localisation of the signal as they alter the local magnetic field experienced by the nuclei and hence changing their precession frequency (f) in a controllable way according to the Larmor equation:

$$f = \frac{\gamma}{2\pi} B_0 \quad (1.3)$$

The transmit coil is responsible for generating a temporary perpendicular magnetic field (B_1) that excites the nuclei, creating rotating magnetisation in the transverse plane, which can be detected by the receive coil as the nuclei relax. Often, transmit and receive coils are combined into one transmit/receive (T/R) coil [1, 5]. The Bloch equation under on-resonance conditions and the rotating-wave approximation, with $B_1 \ll B_0$ yields the duration (t) of the excitation pulse for a given flip angle (α), the angle that the net magnetisation is rotated away from B_0 [1, 6, 7]:

$$\alpha = \gamma B_1 t \quad (1.4)$$

The longitudinal relaxation time (T_1) of the tissue is the time it takes for the net magnetisation of the nuclei to relax from 0% to 63% of its equilibrium magnitude due to spin-lattice interactions. The effective transverse relaxation time (T_2^*) is the time it takes for the precessing spins to dephase and reduce the transverse magnetisation to its 37% maximum amplitude. T_2^* accounts for the irreversible dephasing due to spin-spin interactions (T_2) and the reversible dephasing due to B_0 inhomogeneities (T_2'). T_1 and T_2 are characteristic of each tissue, making it possible to gain information about the sample composition and generate contrast for the MRI.

1.2 Hyperpolarised ^{13}C MRI

MRI sensitivity is normally limited by the Boltzmann distribution of higher and lower energy spin states. Even at high B_0 , the population difference is very small, and a high abundance of the target nuclei is required to obtain a detectable signal. This limitation of the Boltzmann distribution still stands for low abundance nuclei, such

as ^{13}C with an abundance of approximately 1%. In this case, the tracers, such as $[1-^{13}\text{C}]\text{urea}$, but most commonly $[1-^{13}\text{C}]\text{pyruvate}$, are hyperpolarised to achieve detectable signal intensity (Figure 1.1) [8]. Dissolution dynamic nuclear polarisation (d-DNP) is a technique that first polarises unpaired electrons of free radicals in a high magnetic field at cryogenic temperatures, then the electrons are excited by microwave radiation and through coupling the ^{13}C tracer is polarised beyond its thermal equilibrium level [2,4,9]. Dissolving the tracer so that it will be injectable reduces the polarisation to about 70% after which the polarisation decays back to thermal levels in minutes [2,10]. Magnetic resonance spectroscopy is used to capture the spectral and spatial information of the highly transient signal of hyperpolarised sample, enabling metabolic imaging and aiding early detection of disease and treatment response as metabolic changes can precede structural changes [9,11,12].

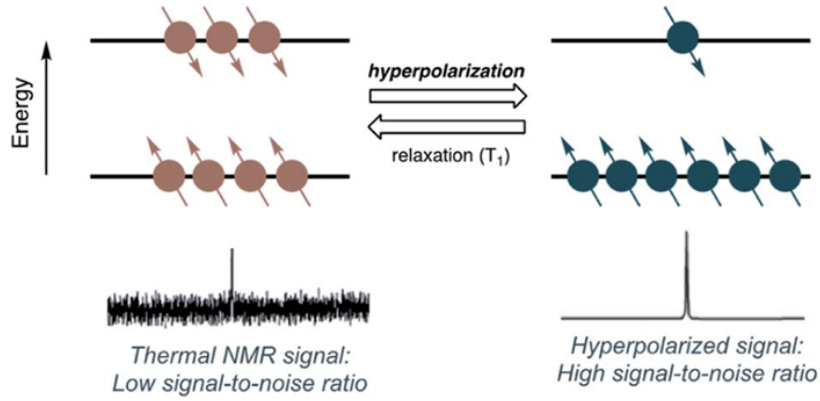


Figure 1.1: Hyperpolarisation of the nuclear-spin population to enhance signal. Reproduced from Park et al. [4] under a CC BY 3.0 licence.

1.3 Polarimetry

The quantification of polarisation is a crucial technique for hyperpolarised tracer evaluation and optimisation. The longitudinal magnetisation, and the integral of the sample spectral peak (S_{sample}) is directly proportional to the polarisation. By measuring S_{sample} , a relative value for the sample polarisation can be obtained [13]. In order to measure the absolute polarisation of the sample according to Equation (1.5), S_{sample} needs to be scaled according to the signal (S_{ref}) of a reference material with known polarisation [14]. S_{sample} is also directly proportional to sample molar concentration (C_{sample}), its effective detection volume (V_{sample}) and its gyromagnetic ratio (γ_{sample}). When the sample itself is used as its own reference at thermal equilibrium,

these scaling factors become identical and the relation simplifies to Equation (1.6). The detection of the thermal signal of a low concentration sample can be challenging that is why a different reference material can be advantageous [14].

$$P_{\text{sample}} = P_{\text{ref}} \frac{S_{\text{sample}} V_{\text{ref}} C_{\text{ref}} \gamma_{\text{ref}}}{S_{\text{ref}} V_{\text{sample}} C_{\text{sample}} \gamma_{\text{sample}}} \quad (1.5)$$

$$P_{\text{sample}} = P_{\text{ref}} \frac{S_{\text{sample}}}{S_{\text{ref}}} \quad (1.6)$$

d-DNP systems are capable of polarimetry, but most of them can only perform relative solid-state ^{13}C polarimetry prior to dissolution, such as the GE SPINlab (GE HealthCare, USA) used at the Oxford Centre for Clinical Magnetic Resonance Research (OCMR). As the polarisation of the sample can reduce during dissolution, liquid state polarimetry is key during multi-centre studies for standardisation and quality control to understand downstream image quality [2, 9]. Alternatively, MRI scanners can perform liquid-state polarimetry [15], but that is an inefficient way to do so because of scheduling issues, high running cost of MRI scanners and unavailability for quality control during a clinical scan.

1.4 Benchtop NMRs

Benchtop NMRs typically differ from their full-sized counterparts in their lack of gradient coils and cryogenic cooling, hence achieving their compact size, affordability and their lower B_0 [16, 17]. Without gradient coils, they are unable to perform signal localisation, while without cryogenic cooling, they cannot use superconducting magnets. They generate the B_0 either with permanent magnets, electromagnetic coils, or use the Earth's magnetic field. Other components of benchtop NMRs are usually comparable to a full-sized ones (Figure 1.2). The NMR console orchestrates the shim control, temperature control, pulse generation, acquisition and data processing [18]. The transmit signal amplifier and the power supply can also be considered as part of the console. The usual setup for the NMR probe is a tuned and matched radio frequency (RF) coil, with pre-amplifier and switching between transmit and receive modes. T/R switching is necessary to isolate the sensitive pre-amplifier from the high power output during excitation [19]. The lower main field for benchtop NMRs also makes it possible to use optically pumped magnetometers (OPMs) as the NMR probe. OPMs are highly sensitive to low magnetic fields and feature an integrated

pre-amplifier, however using them as an NMR probe requires a separate transmit coil [13, 20].

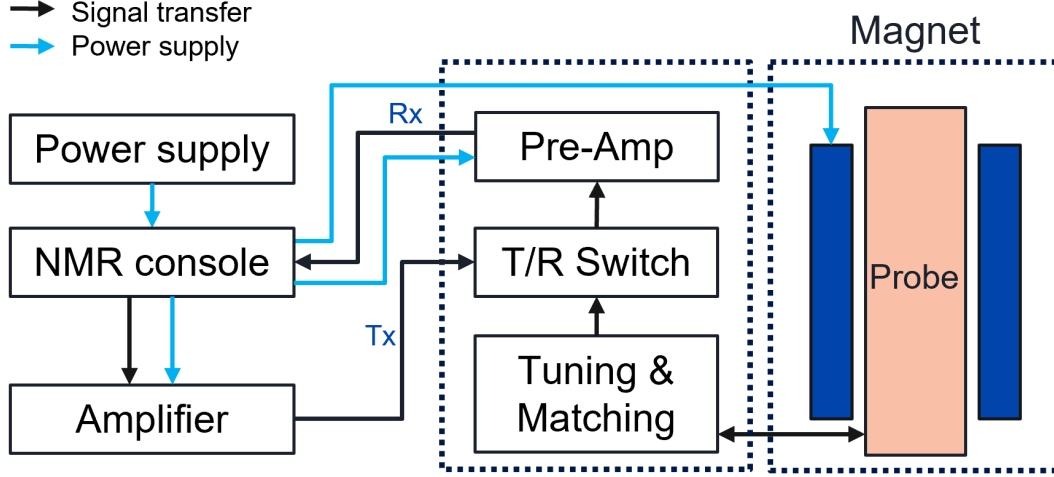


Figure 1.2: General NMR schematic.

For example, in Mouloudakis et al. [13] a high-sensitivity miniaturised ^{87}Rb -vapour magnetometer was used for non-destructively measure the polarisation of hyperpolarised ^{13}C . In this setup, dynamical decoupling pulse sequences were used to repeatedly invert the sample polarisation. The magnetometer was capable of measuring longitudinal magnetisation, which eliminated the need to generate transverse magnetisation with destructive flip angles in favour of 180° pulse. The inversions facilitated a more robust way of separating the static main magnetic field and acquiring the signal in contrast to subtracting thermal signal after decay. This technique has been used to continuously and non-destructively observe hyperpolarised spin dynamics [20] and proposed as a quality control procedure for hyperpolarised tracers.

Another custom-made NMR spectrometer, using a T/R coil and electromagnet in a four-coil Barker arrangement for B_0 was described by Chowdhury et al. [14]. They perform absolute ex situ polarimetry on hyperpolarised ^{13}C . Their spectrometer can also be incorporated into a ^{129}Xe hyperpolariser to perform in situ ^{129}Xe polarimetry [21]. Its NMR console was built from affordable components, including a pre-amplifier and a T/R switch triggered by a bias current. An integrated chip containing two PIN switch diodes connected in series was used for the T/R switch [14, Supporting Information Figure S1].

An ultra-low field benchtop NMR using a commercially available multifunctional input/output device was proposed by Bryden et al. and was able to acquire both

^1H and ^{129}Xe signals at mT range [22]. Potentially, it could also be used for ^{13}C polarimetry. It bypasses the frequency limitation of the ADC by down-converting the frequency of the input signal.

Commercially available benchtop devices for ^1H spectroscopy such as the MQC-R (Oxford Instruments, UK) are capable of ^{13}C polarimetry, but they are costly and over-optimised to solve this problem [23,24]. Alongside its fundamental capability of relaxometry for quality assurance and quality control of small samples, it was also designed to be customisable for research purposes. Its probe is a interchangeable T/R coil with available effective detection volumes of 1, 7, and 14 mL. It has a temperature-regulated Neodymium permanent magnet providing a B_0 of 0.55 T corresponding to a ^1H resonance frequency of 23.4 MHz.

1.5 NMRduino

NMRduino is an open-source project about the development of a low-complexity NMR console. Its layout can be seen in Figure 1.3. NMRduino was designed around inexpensive components and with accessibility, robustness and customisability in mind. Its small size (8.5×5.4 cm) also makes it highly portable. The main processing component of NMRduino is the Teensy 4.1 (SparkFun Electronics, USA) microcontroller unit (MCU), which is responsible for the communication with the personal computer (PC), pulse programming, and the synchronisation of the components during pulse sequence generation. Because the Teensy 4.1 can only generate a positive low power (3.3 V, 4 mA) digital signal, it requires a series of components to form the final RF pulse, drive peripheral components with alternating current (AC) or bipolar direct current (DC). NMRduino can generate three bipolar DC outputs by channelling an external power supply through H-bridges (bidirectional switches), which polarity is controlled by the MCU. To generate two true AC signals, the digital output signals first have to be converted by a digital-to-analogue converter (DAC) (Microchip Tech., USA; MCP4822-E/SN), then amplified by an operational amplifier (STMicroelectronics, Switzerland; L272D), and finally the H-bridges, DRV8838 and DRV8835 (Pololu, USA; 2990 and 2135) generate the negative amplitudes of the wave. For the two analogue signal inputs the NMRduino has no pre-amp but has an analogue-to-digital converter (ADC) (Analog Devices, USA; LTC1859CG#PBF) to digitize the input signal and make it readable for the MCU.

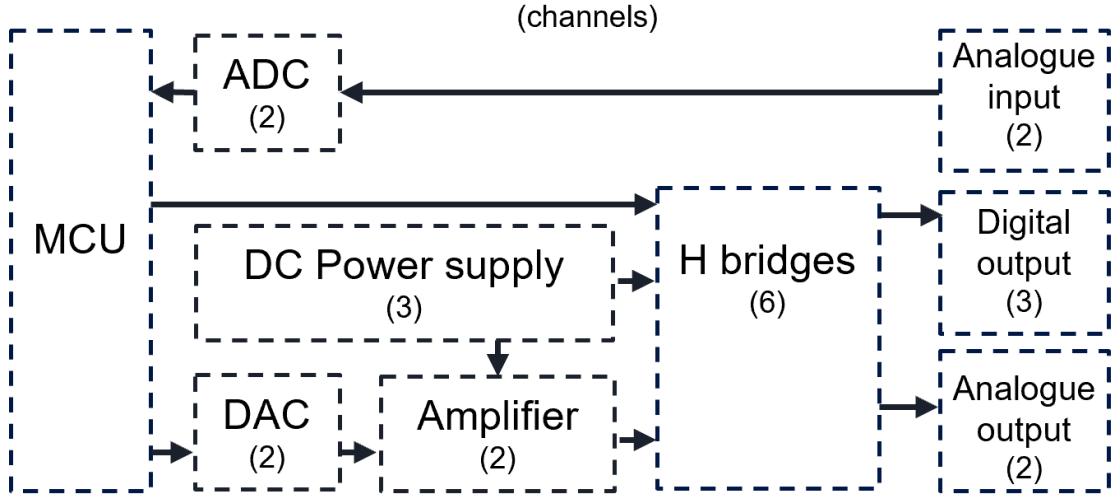


Figure 1.3: NMRduino layout. Number of channels that a unit can process is denoted in brackets. Adapted from Tayler et al. [25] under a CC BY 4.0 licence.

With the digital outputs, the NMRduino can drive motors, the main magnetic field, power pre-amplifier or generate T/R switch triggers. Analogue outputs are for driving the transmit coil or generating a rotating magnetic field.

The maximum supply voltage for each channel is limited by the H-bridges to be 11 V, however in the NMRduino schematic the maximum input voltage is recommended to be 10.5 V [26]. The MCU is powered by the PC via its universal serial bus (USB). The continuous DC output current is also limited by the H-bridges to 1.2 A for a single channel, or 2.4 A when two channels are combined. While L272D limits the analogue output current to 1 A.

Firmware is required on the MCU side that interprets and executes the pulse sequences. A user interface (UI) makes the communication possible with the MCU from the PC side, through which the input data can be received and the pulse sequences can be programmed and sent to the MCU.

The sampling rate of the used ADC was 100 ksps, which according to the Nyquist theorem could sufficiently sample a 50 kHz signal, corresponding to a B_0 of 4.67 mT for ^{13}C . According to guidelines it is recommended to keep the signal below 80% of the Nyquist limit to practically reproduce the signal amplitude in the frequency domain when using an anti-aliasing filter before the analogue-to-digital conversion [27]. A tuned probe acts as a band pass filter, and when set to a low enough frequency can negate the need for an anti-aliasing filter.

To increase the achievable resonance frequency of the NMR, it is possible to use the inbuilt ADC in the Teensy microcontroller, which has a sampling rate of 1 Msps or alternatively use MAX1300 (Analog Devices, USA) with 115 ksps as the ADC. The printed circuit board (PCB) available in Table A.2 was designed for LTC1859, so it would need to be redesigned to accommodate an alternative ADC [25]. For hyperpolarised acquisition, increasing the B_0 field would not improve the signal as the sample is already polarised beyond the Boltzmann distribution. For shielding purposes it is also not necessarily beneficial to increase the tuned frequency, as there could possibly be more instrument interfering with the signal at higher ranges. However, a higher frequency would increase the thermal signal.

1.6 Tuning and matching

Tuning of a T/R coil to the Larmor frequency of the sample is necessary for it to excite the nuclei and also to be able to detect the transverse magnetisation. Matching the impedance of an RF component (Z_L) to the source impedance (Z_0) is needed as part of the signal can reflect from the connection interface, resulting in power loss if their impedances are drastically different at the frequency of the signal. RF components usually made with the standard purely resistive $50\ \Omega$ impedance to be compatible with each other and minimise power loss due to reflection.

There are several ways of building a circuit that simultaneously provides tuning and matching [7, 14, 28–31]. Tuning stubs (short sections of transmission lines) are perhaps the simplest matching circuit, but are not ideal for sub MHz signals due to the long cable they would require [7, 28].

A simple and practical method for impedance matching low frequency RF coils is the L-section matching network, which can also tune the coil to the desired frequency. It has two possible configurations, if the load is purely resistive and its AC resistance (R_L) is higher than the resistance of the source, Figure 1.4a is recommended, otherwise Figure 1.4b [28]. Because of the reactance of a coil (X_L) is positive, both layouts are possible, but require significantly different series reactance value (X) and shunt susceptance (B). The series element moves the impedance of the load along a constant resistance circle, while the shunt element changes the susceptance. The needed reactance values for the layout in Figure 1.4a can be calculated from Equations (1.7) and (1.8), and for Figure 1.4b can be calculated using Equations (1.9) and (1.10) [28, 31]. Because of the quadratic nature of both sets of equations, each produces two valid solutions.

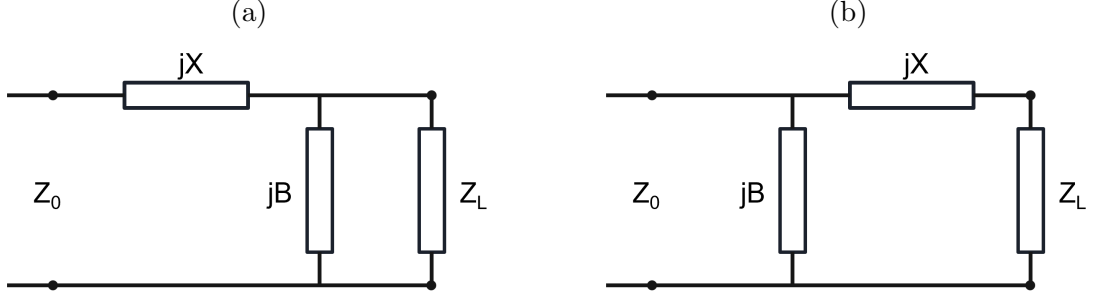


Figure 1.4: L-section tuning and matching circuit: (a) $R_L > Z_0$, (b) $R_L < Z_0$. Adapted from Steer et al. [31] under a CC BY 4.0 licence.

$$B = \frac{X_L \pm \sqrt{\frac{R_L}{Z_0} \sqrt{R_L^2 + X_L^2} - Z_0 R_L}}{R_L^2 + X_L^2} \quad (1.7)$$

$$X = \frac{1}{B} + \frac{X_L Z_0}{R_L} - \frac{Z_0}{B R_L} \quad (1.8)$$

$$X = \pm \sqrt{R_L(Z_0 - R_L)} - X_L \quad (1.9)$$

$$B = \pm \frac{\sqrt{(Z_0 - R_L)/R_L}}{Z_0} \quad (1.10)$$

1.7 Quality factor

The intrinsic quality factor (Q_0), the ratio of the stored and dissipated energy, was calculated with Equation (1.11). The definition of unloaded quality factor (Q_{unloaded}) is different between RF engineering and the field of NMR probe design. In this thesis the NMR definition was adopted, which defines Q_{unloaded} as the matched quality factor of the coil without a sample and calculated with Equation (1.12) [28, 30, 32]. f_0 is the resonance frequency and Δf is the bandwidth between the two frequencies with -3 dB reflection coefficient [30]. The second term in Equation (1.12) is the external circuit contribution to Q_{unloaded} and is determined by how well the system is matched. The coefficient of coupling g is the ratio between source impedance and matched AC resistance of the coil. In case of a critically coupling i.e. perfect matching ($Z_0 = R_L$), $g = 1$, making Q_{unloaded} half of Q_0 as the external circuit contributes an equal resistance to the system [28]. DC resistance was a good approximation for the AC resistance in a 0.25 mm copper wire at the 10 kHz frequency range, as the skin

effect was negligible due to the skin depth being 0.66 mm [32]. The quality factor of the coil when it is loaded with a sample (Q_{loaded}) can be calculated from $\frac{f_0}{\Delta f}$, but in this case both Δf and f_0 are the characteristics of the loaded coil. The ratio of unloaded to loaded quality factors (Q_{ratio}) is recommended to be greater than 2 to improve signal-to-noise ratio (SNR) [30].

$$Q_0 = \frac{X_L}{R_L} \quad (1.11)$$

$$Q_{\text{unloaded}} = \left(\frac{1}{Q_0} + \frac{1}{gQ_0} \right)^{-1} = \frac{f_0}{\Delta f} \quad (1.12)$$

1.8 Overview

The aim of this work was to design a cost efficient benchtop NMR capable of absolute liquid-state ^{13}C polarimetry. Currently, at the OCMR, a clinical 3 T MRI scanner is used for liquid-state polarimetry. The benchtop NMR developed during this project (Figure 1.5) was meant to provide an efficient alternative to the MRI scanner for polarisation measurements. Its NMR console was based on the open-source NMRduino platform [25]. NMR devices have been made using the same platform but with less accessible components, where a magnetometer was used as an NMR probe [7, 13, 20]. Different simulation techniques, Ansys Maxwell and High-Frequency Structure Simulator (HFSS) were compared and used to aid design and construction of a Helmholtz coil that generated the main magnetic field and a solenoid for a T/R NMR probe. An interface box was made for the L-section tuning and matching circuit and for a future T/R switch and pre-amplifier. The custom-made structural parts of the NMR, such as the enclosures, sample holder and coil winding forms were 3D printed. In this thesis, the development towards such a device and its components is described and evaluated.

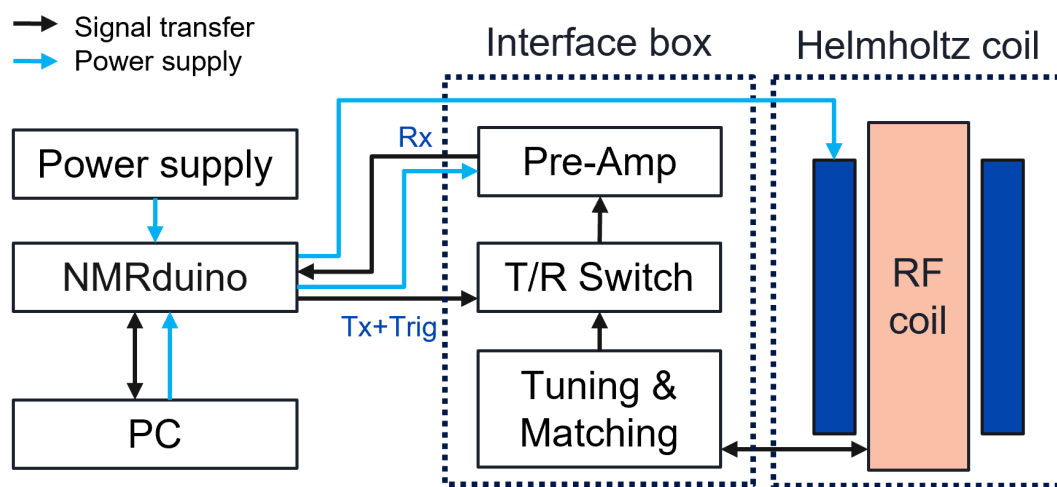


Figure 1.5: NMR schematic with NMRduino.

Chapter 2

Materials and Methods

2.1 Construction of NMRduino

All the components used for the construction of the designed benchtop NMR polarimeter are listed in Appendix A, including the NMRduino, parts of the interface box and the coils.

Initially, surface-mount devices, such as the ADC, DAC, amplifier and capacitors, were soldered to the PCB (Figure 2.1b). Some optional soldering pads were left unused, such as the one that would fit a 1 or 10 μF capacitor between the V_{CC} pin of the DRV3538 motor driver and ground, or the pads above the L272D amplifier that could be used to smooth the signals of analogue output channels.

The header pins and connection ports were then soldered to the board (Figure 2.1c). The outer shell of a Bayonet Neill–Concelman (BNC) port partially covers one of the soldering pads of a DRV8835 (Figure 2.1d). As both are intended to be connected to ground, it does not cause any issues, but it would still be better to use smaller-footprint BNC ports that fit better on the PCB and also leave more room for the screws securing the PCB to its case.

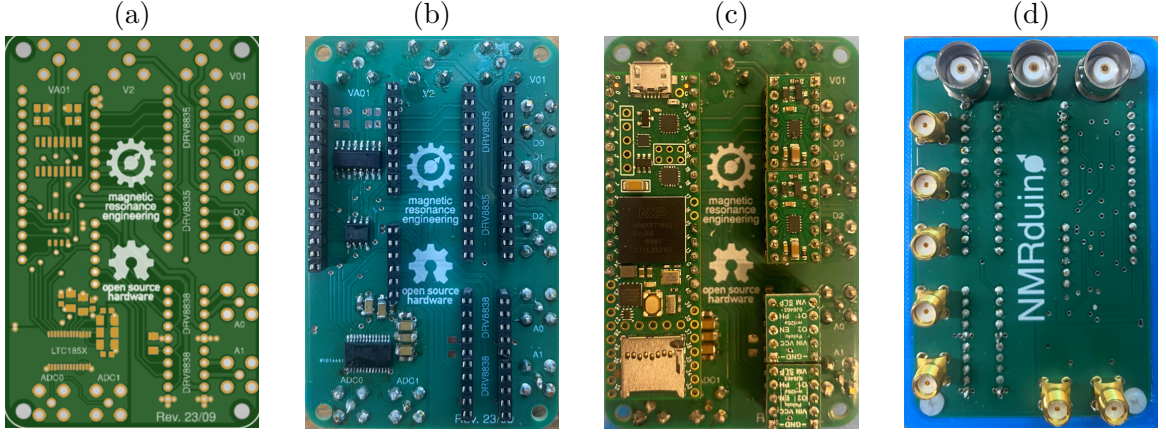


Figure 2.1: NMRduino PCB: (a) bottom layer illustration, (b) bottom layer with amplifier, analogue to digital converter and digital to analogue converter, (c) bottom layer with carrier board attached, (d) top layer.

2.2 Simulations

The simulations were performed in the electromagnetic simulation software, Ansys (Ansys inc., USA; v2026 R1), with Maxwell solver, which is recommended for DC and sub MHz frequencies and HFSS recommended for MHz and upward frequencies.

Maxwell (Magnetostatic) was used for the simulation of the Helmholtz coil, primarily because of its ability to simulate DC excitation, but also because its ability to implicitly model the coil just by its cylindrical shape and the specified number of turns made its use more practical.

The T/R solenoid was simulated with HFSS and Maxwell, but because AC excitation was applied in this case, the AC eddy current (T-Omega) solution type was chosen for the Maxwell simulations. The solenoid was modelled with a winding thickness of its actual wire diameter 0.25 mm and also 0.163 mm. The 0.163 mm was an adjusted value that accounts for the discrepancy between the cross-sectional area of the cylinder and the winding. Then, the simulated characteristics of the coil was imported into circuit design, where the reflection coefficient (S_{11}) and forward transmission coefficient (S_{21}) could be deduced. B_1 was simulated using this method. HFSS can also plot the B_1 indirectly, using the field calculator to deduce it from the magnetic field strength (H), but Maxwell was chosen due to the more direct way of controlling the driving current. All simulations were done at 22 °C, except the

Maxwell simulation with the adjusted winding thickness, which was also repeated at the maximum operating temperature 55 °C.

For the two Maxwell simulations, the advanced analysis setup was set to have a maximum of 20 iterations out of which the energy percentage error (mesh elements/total energy) required to be below the 0.1% threshold for two consecutive iterations for the simulation to conclude.

For the HFSS simulation of the solenoid coil, a 266-turn helix needed to be explicitly modelled, which significantly increased the computational demand of the simulation. A single excitation port was implemented, which implies that the other end of the coil is grounded. The HFSS simulation was done with automatic analysis setup in order to ease its computational demand. The HFSS simulation was also repeated with two excitation ports connected by a perfect conductor reference plane in order to be able to create two pins and hence simulate S_{21} , later in circuit design.

2.3 3D designing and printing

The 3D printed parts were designed in Autodesk Fusion (Autodesk, USA; v2.0.19440) to be printable without support material. PrusaSlicer (Prusa Research, Czech Republic; v2.9.6) was used to process the 3D models into stereolithography (STL) files with a 0.15 mm BALANCED print profile, to prepare the 3D models for printing in polylactic acid (PLA) on a Prusa CORE One+ (Prusa Research, Czech Republic) 3D printer. A 0.3 mm diametric clearance was applied to the models, which, together with their corresponding STL files were made available in a GitHub repository that can be found in Table A.2.

Based on the deflection temperature of PLA, determined by ISO 75 testing standard of a 80 mm long, 10 mm wide, and 4 mm thick rod under 0.45 MPa, the maximum operating temperature was set to 55 °C [33]. It was not a hard limit because the frame has no load-bearing function and hence not experiencing any significant stress, perhaps other than the tension of the coil winding.

2.3.1 Enclosures

The original model for the NMRduino case (Figure 2.2a) is openly available as part of the NMRduino project [25] and was adapted to fit the Subminiature version A (SMA) and BNC connector pins with a slightly larger footprint than it was designed for. Its model needs to be split horizontally along the top of the micro-USB port to eliminate the overhang and make it printable without support material. It was

designed to accommodate threaded inserts that were used to secure the PCB to the case with nylon screws.

The enclosure for the interface box (Figure 2.2b) had ledges on the sides and was also designed with legs to elevate the stripboard and to provide a way to impermanently secure the board by adding 1 mm inner diameter rubber O-rings to the legs once the board was in place. As the enclosure was non-magnetic, the circuit board did not need to be electrically insulated from it, but the separation improved the cooling of the board. The front panel of the enclosure slid into its slot and was eventually secured in place by the lid, which was attached with screws.

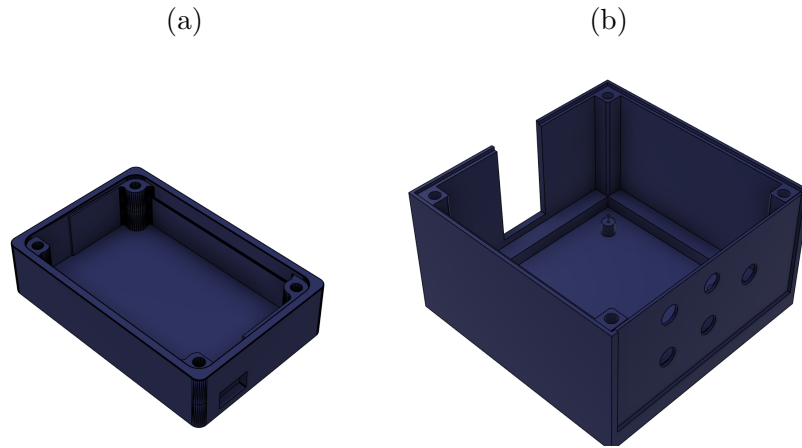


Figure 2.2: Illustration of (a) the case for NMRduino ($2.2 \times 6 \times 9$ mm)(b) and the enclosure of the interface box ($90 \times 90 \times 50$ mm).

2.3.2 Sample holder

2.3.2.1 Sample holder with Helmholtz coil

The sample holder of the benchtop NMR (Figure 2.3) was designed around a 10 mL Plastipak syringe with Luer-Lok (BD, USA; Ref. 305959). This syringe has a barrel length of 8.4 cm, of which the rubber seal reducing the effective length of the barrel to 7 cm (excluding the tip). The effective length of the barrel, together with its diameter of 1.6 cm and a larger than usual clearance, determined the length and inner diameter of the solenoid coil form to be 8×1.7 cm. The sample holder was also made to accommodate the 10 mL eccentric-tip syringe (Medicina Ltd, UK; Ref. IVS10E), which has a barrel length of 7.5 cm and a similar 1.6 cm diameter.

The coil form of the T/R solenoid was printed in two pieces, which were joined with Loctite cyanoacrylate adhesive (Henkel, Germany; Ref. 853188). The wound

coil was secured to the frame with non-magnetic threaded inserts and screws. That makes it possible to easily remove and change the solenoid.

The magnetic flux density of the Helmholtz coil (B_0) at the centre is inversely proportional to its radius according to the equation derived from the Biot-Savart law [34], where μ_0 is the permeability of free space, N is the number of turns, I is the current, and R is coil radius:

$$B_0 = 0.8^{1.5} \frac{\mu_0 N I}{R} \quad (2.1)$$

Consequently, the radius was chosen to be as small as reasonably achievable to minimise its resistance and hence avoid exceeding the voltage limit of the PCB output port (10 V). The radius of the Helmholtz coil must be equal to the distance between the coil pair, therefore it was set to 3 cm with each coil having a length of 0.5 cm, leaving a sufficiently large gap where the T/R solenoid can still be fitted. Each of the Helmholtz coils was printed in two parts that could be screwed together.

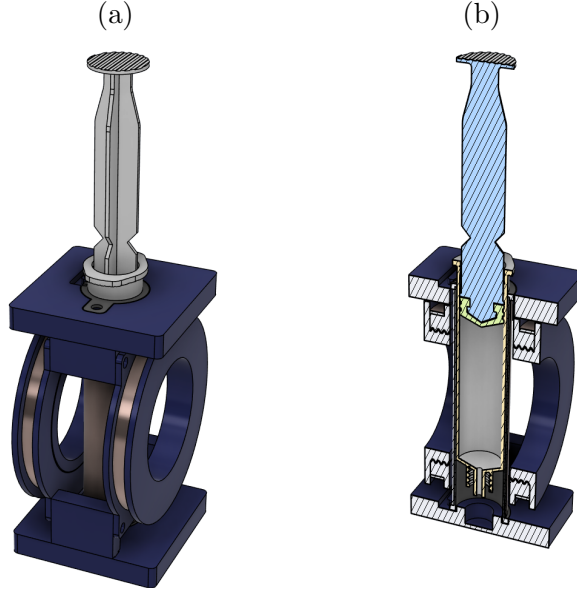


Figure 2.3: (a) Illustration of the sample holder (Helmholtz coil pair providing the main magnetic field and the transmit and receive solenoid), (b) Sample holder cross section.

2.3.2.2 Sample holder with permanent magnets

The option of using permanent magnets for B_0 were also considered. A design of the sample holder was made for a pair of $5 \times 5 \times 2.5$ cm, grade N52 permanent neodymium magnets (Unimagnet, Czech Republic; Ref. KV-50-50-25-N52N) with a residual magnetism of 1.45 T and pull force of 138 kg. The plan was to make the sample holder out of wood instead of PLA for mechanical strength and to use aluminium plates to keep the magnets apart. The attraction between the two magnets would keep them in place, but a steel sheet could be wrapped around the side of the magnets to secure them and also act as a yoke, guiding the return path of the magnetic flux and increasing B_0 . Because in this setup, B_0 would already be too high without a yoke, a simple 3D printed notch could be added at the back instead of a yoke.

The drawback of using permanent magnets was distance by which they needed to be separated in order to generate a low enough magnetic field to set a resonance frequency for ^{13}C below 50 kHz, so that it would be within the Nyquist limit of the analogue to digital converter used. The two magnets would need an impractical centre-to-centre separation of 34 cm, in which case the homogeneity of the field would also worsen. This setup was simulated with Maxwell (Magnetostatic) and the results are presented in Appendix B. Hence, it was more practical and controllable to generate the magnetic field in the mT range with an electromagnet.

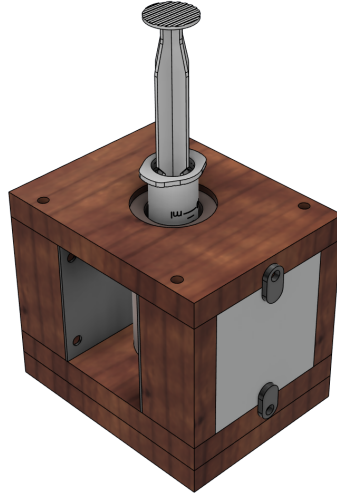


Figure 2.4: Illustration of sample holder with permanent magnets.

2.4 Construction and testing of the coils

Both the solenoid and the Helmholtz coil were wound with a single core 0.25 mm diameter copper wire. An enamel layer with an outer diameter of 0.3 mm kept the turns electrically insulated from each other. This wire had a maximum operating temperature of 155 °C and a nominal resistance per unit length of 0.351 Ω/m at 22 °C, while for different temperatures Equation (2.2) was used to calculate the wire resistance where the temperature coefficient of resistance (α) of copper is 0.00393 1/°C.

$$R = R_{\text{ref}}(1 + \alpha(T - T_{\text{ref}})) \quad (2.2)$$

2.4.1 Transmit/receive solenoid

Limited by the high resistance of the wire, the aim was to wound 260 turns in the first layer of the RF solenoid and a remaining 6 in the second layer (Figure 2.5). This way the winding starts and ends on the same side of the coil, making it easily changeable inside the sample holder. Keeping it essentially a single layer coil also made it more comparable to the single layer HFSS simulation. The wire length of the RF solenoid was 15.4 m. The final dimensions of the coil were an average diameter of 18.45 mm and 80 mm length. The turns were attached to the form with two axial lines of cyanoacrylates glue. The B_1 field was measured with a Gauss meter (PCE Instruments UK Ltd, UK; PCE-MFM 3500) only at the top of the solenoid, because the probe can only measure the magnetic flux density perpendicular to its surface. Equation (2.3) was used for the calculation of the theoretical B_1 , where l is the coil length.



Figure 2.5: Constructed T/R solenoid.

$$B_1 = \frac{\mu_0 N I}{\sqrt{4R^2 + l^2}} \quad (2.3)$$

To measure the inductance (L) of the T/R solenoid, it was connected to the network analyser, after a 0.2 nF multilayer ceramic capacitor was connected in parallel between the two terminals of the coil (Figure 2.6). The inductance test of the solenoid was done by measuring S_{11} and S_{21} using an 8712ET vector network analyser (VNA) (Keysight, USA. Frequency range: 300 kHz-1.3 GHz). From the measured resonance peak, the inductance could be calculated using the Thomson formula (Equation (2.4)), where C is the capacitance. For comparison the inductance of the solenoid was deduced from simulations in Ansys and also calculated from its dimensions from the Wheeler formula (Equation (2.5)) and the ideal equation of inductance (Equation (2.6)), where A is the cross-section of the coil. The Wheeler formula for finite length solenoid calculates the inductance with 1% error when the length-to-radius ratio is higher than 0.8 [35].

$$f = \frac{1}{2\pi\sqrt{LC}} \quad (2.4)$$

$$L = \frac{10^{-6}}{0.0254} \frac{R^2 N^2}{9R + 10l} \quad (2.5)$$

$$L = \frac{\mu_0 N^2 A}{l} \quad (2.6)$$

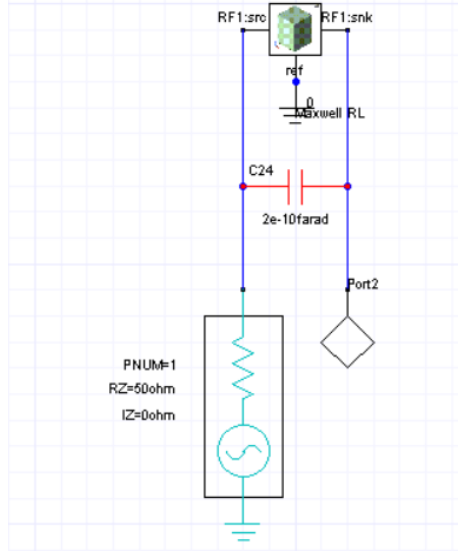


Figure 2.6: Inductance measurement circuit layout in transmission mode. Maxwell RL is the T/R solenoid, C24 is the test capacitor, and PNUM=1 (port 1) and port 2 are the ports of the VNA.

2.4.2 Helmholtz coil

The Helmholtz coil was built with 67 turns in 6 layers (Figure 2.7). As there were no grooves on the form to guide the wire, it was a random-wound coil. The core diameter of the form was 60 mm, and because of the random winding the average radius of the coil was less than the expected value and was measured to be approximately 30.6 mm. For each Helmholtz coil 12.6 m of wire was required. One end of each coil was connected, so that it could be driven from the same power supply ensuring a uniform current and improving homogeneity. Then the B_0 was also measured with a Gauss meter in the inside of the attached T/R solenoid between 3 and 5 cm of its height. Because a thermal polarisation acquisition is 15 minutes with the 3 T scanner, the coil temperature was measured for the same amount of time using a handheld multimeter (RS Components, UK; Ref. RS-14) to choose a baseline driving current (0.6 A) for tuning and matching that did not exceeding the PLA deflection temperature. The temperature was measured on the surface of the wire, but due to the small winding thickness of 1 mm it was assumed to be the same as the face of the winding guide. The baseline driving current was eventually fine tuned to match the actual resonance of the solenoid.

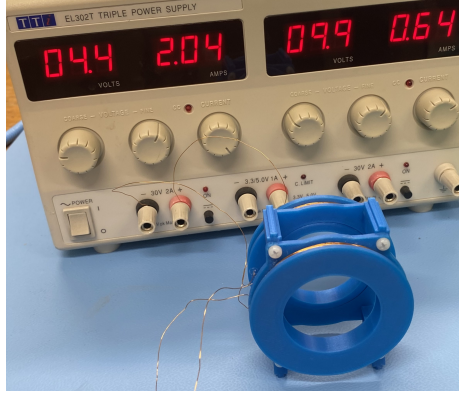


Figure 2.7: Constructed Helmholtz coil.

2.4.3 Tuning and matching

The resistance at room temperature (22 °C) was used to calculate the required tuning and matching reactances from Equations (1.7) to (1.10) to achieve the target baseline frequency of 13.6 kHz. The negative-sign solutions of Equations (1.7) and (1.8) was implemented, because they provided the most balanced setup compared to the other options. It still yielded a relatively large 204 nF series and 323 nF shunt capacitance (Table 2.1).

Table 2.1: Component values for the two tuning and matching layout in Figure 1.4.

	Series	Shunt
Circuit a-	204 nF	323 nF
Circuit a+	669 μ H	555 nF
Circuit b-	283 nF	213 μ H
Circuit b+	643 nF	1270 nF

Because of the high capacitance needed, polypropylene film capacitors with 5% tolerance were chosen for impedance matching. They were also preferred because of their adequate DC voltage rating of 630 V. The leads of the capacitors were tested with a permanent magnet and turned out to be magnetic. The leads were trimmed after soldering and the capacitors were placed outside of the main field in order to minimise interference with B_0 . At this high capacitance, it was not feasible to use trimmer capacitors at this range of capacitance, so the fine-tuning was done by adjusting the B_0 field. The capacitance was divided into smaller segments, two pieces of 100 nF for the series elements and 220 nF and 100 nF for the shunt elements (Figures 2.8 and 2.9). The final evaluation of the matched coil was done with a

NanoVNA-H (Zeenko, China. Frequency range: 9 kHz-1.5 GHz). The number of scan points of the NanoVNA-H was 101 and it was set to sweep with a frequency step of 0.26 kHz over a 26 kHz range centred on the target resonance frequency of 13.6 kHz, without averaging over time points. Because of the limited amount of scanning points, the -3 dB frequencies, hence the matched Q was calculated from the linearly interpolation of the two frequencies surrounding the -3 dB threshold.

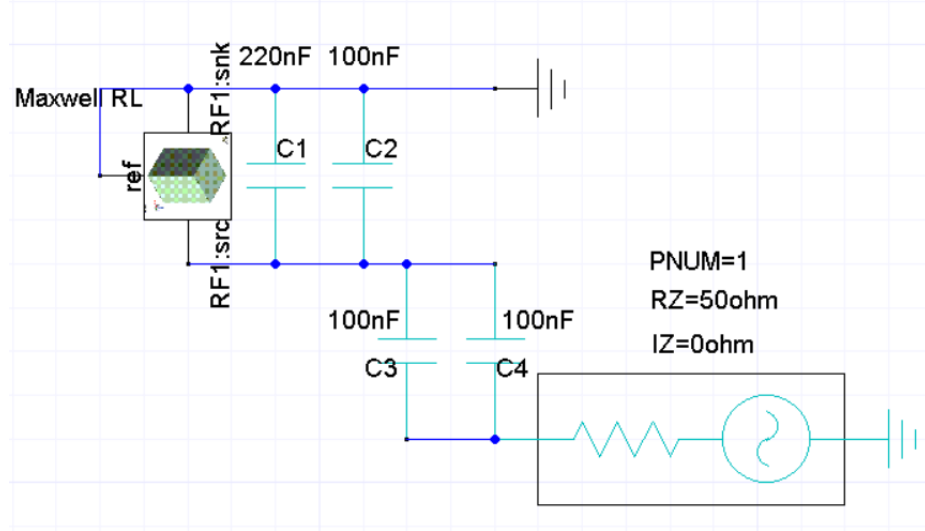


Figure 2.8: L-section tuning and matching circuit layout. Maxwell RL is the T/R solenoid grounded through the outer conductor of the VNA port 1, PNUM=1 is the centre pin of the VNA port 1 and C1-C4 are the polypropylene film capacitors.

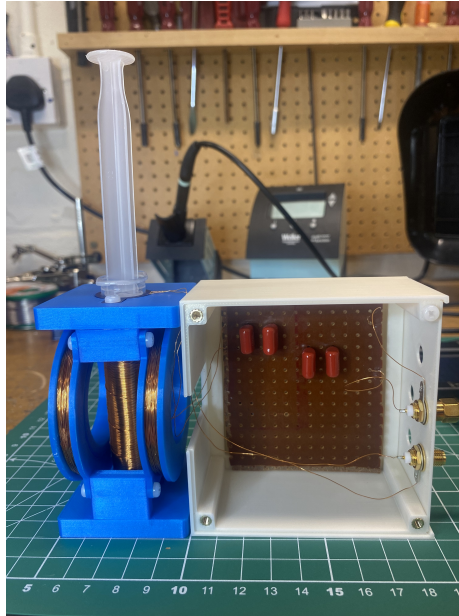


Figure 2.9: Tuning and matching circuit with sample holder.

Chapter 3

Results

3.1 Helmholtz coil

3.1.1 Experimental

The constructed Helmholtz coil, made from a relatively small 0.25 mm diameter wire, had high resistance of 11.2 Ω at 22 °C and dissipated excessive heat, which further increased its resistance and heating. Depending on the current, the coil reached different equilibrium temperatures. Out of the three current levels tested, only the trial with 0.6 A kept the Helmholtz coil below the set maximum operating temperature of 55 °C throughout the 15 minutes (Figure 3.1), in which case coil reached 52 °C and 12.2 Ω resistance at the end. The coil driven by 0.65 and 0.7 A was still increasing in temperature at the end and would presumably reach the glue service temperature (82 °C), before reaching equilibrium [36].

0.6 A was chosen as the baseline driving current for the Helmholtz coil as it kept the coil reasonably within the operating temperature, with some room for fine adjustment. It determined the resonance frequency to be 13.6 kHz for ^{13}C . The median B_0 with 0.6 A, was 1.27 mT between the height of 3 and 5 cm, while the theoretical value was 1.18 mT. Based on the tuned and matched T/R coil, the final driving current and magnetic field of the Helmholtz coil were 0.63 A and 1.32 mT.

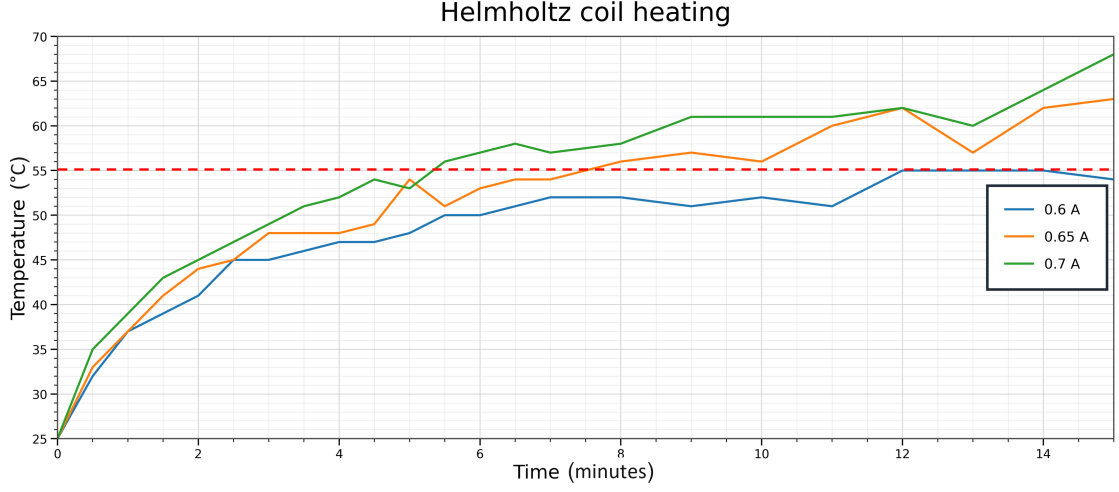


Figure 3.1: Helmholtz coil heating for different driving currents.

3.1.2 Simulation

The resulting heat map of B_0 of the Helmholtz coil from the Maxwell simulation performed at 0.6 A is shown in Figure 3.2a. It provides a qualitative assessment of field homogeneity, while the plot in Figure 3.2b overlay B_0 along the axis of the Helmholtz coil (horizontal midline) and along the RF solenoid axis (vertical midline). The theoretical B_0 in the centre was 1.18 mT compared to the simulated 1.12 mT. The theoretical Q_{unloaded} of the perfectly matched solenoid was calculated to be 2.12 from the resistance at 22 °C and the inductance using the Wheeler formula. Such a Q_{unloaded} predicted Δf to be 6.43 kHz, corresponding to ΔB_0 of 0.6 mT at all resonance frequencies if the inductance of the coil kept unchanged. The solenoid was tuned to the central B_0 , halving the effective ΔB_0 . As the Helmholtz coil was more homogeneous along its axis, the limiting factor of the effective detection volume was the variation of B_0 along the vertical midline. Figure 3.2b shows that ^{13}C can be excited over a 48.5 mm range.

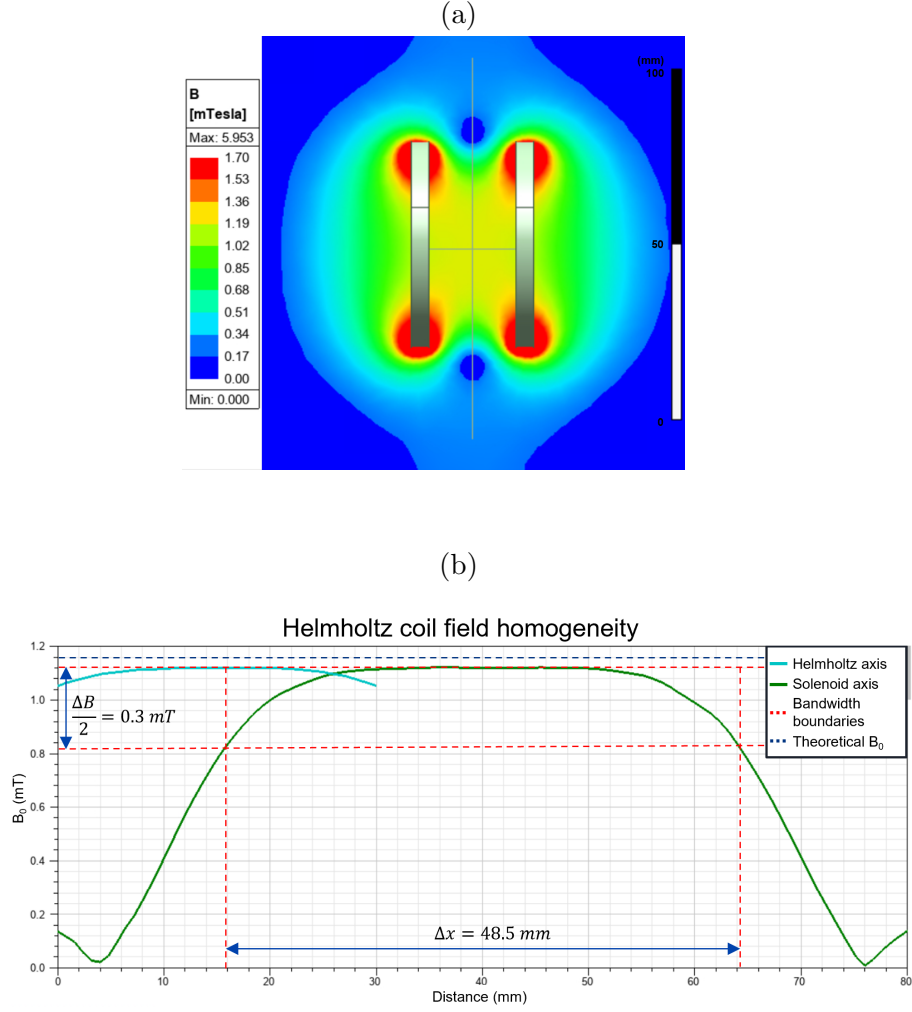


Figure 3.2: Helmholtz coil simulated with 0.6 A (a) magnetic flux density heat map and (b) plot along its Helmholtz coil axis (horizontal midline) and solenoid axis (vertical midline) with overlaid frequency bandwidth calculated from the theoretical matched Q_{unloaded} .

3.2 Transmit/receive solenoid

3.2.1 Experimental and simulation

The Maxwell simulation of the T/R solenoid with the actual winding thickness yielded a field-per-current of 3.85-4.05 mT/A_{peak} over the 48.5 mm long range (35.8-84.3 mm) determined by the homogeneity of B_0 shown in Figure 3.3. The theoretical field-per-current of the solenoid was 4.07 mT/A_{peak} calculated from Equation (2.3), the finite length corrected magnetic flux density. A Gauss meter was used to measure B_1 at the upper end of the solenoid. This measurement position corresponded to 102.5 mm in

the simulated magnetic field plot (Figure 3.2b) because the field is plotted along a line extending beyond the solenoid. The experimentally measured B_1 value at 102.5 mm was 1.57 mT compared to the simulated value of 1.51 mT, supporting the validity of the simulation.

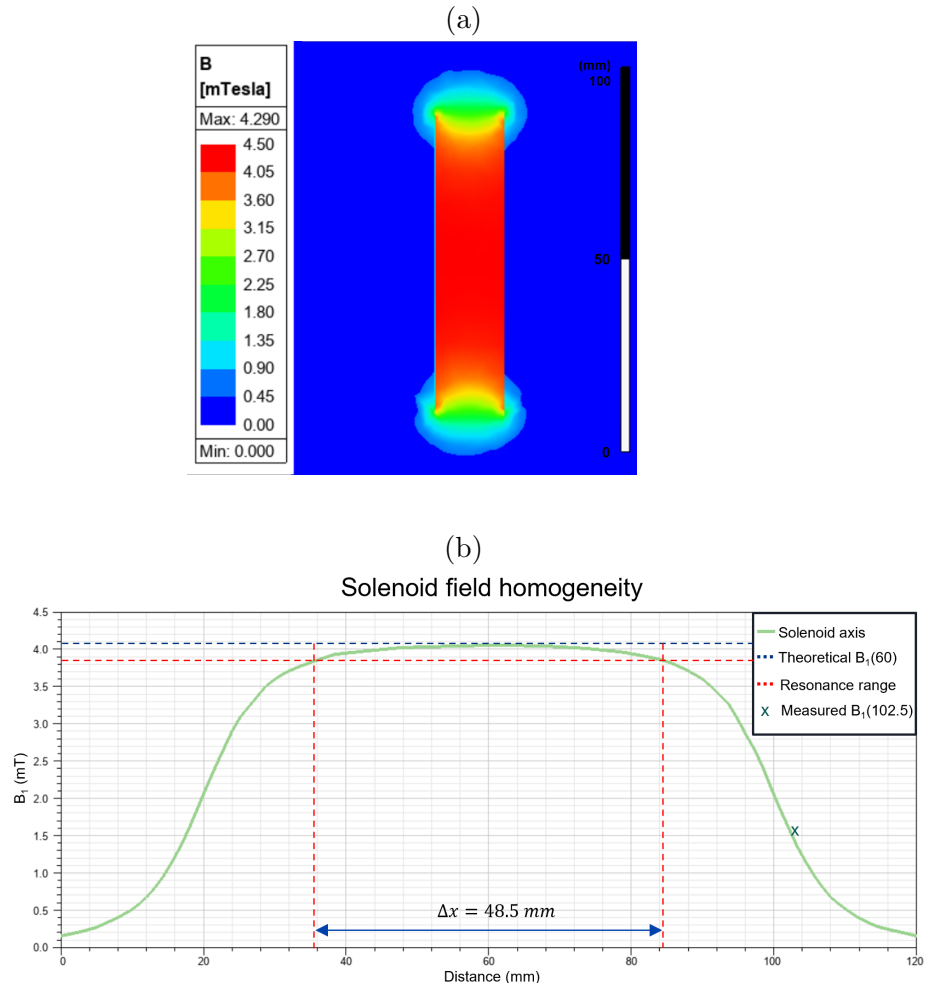


Figure 3.3: Solenoid (a) magnetic flux density simulated heat map and (b) plot along its axis at 1 A.

For the theoretical value of inductance, the result from the Wheeler formula was used as it adjusts for the finite length of the solenoid. The difference in winding thickness did not affect the inductance value between the two Maxwell simulations shown in Table 3.1, both with an approximate -2% relative error. However, it did change the resistance as expected. Using the actual winding thickness (0.25 mm) in the Maxwell simulation resulted in resistance 34.2% lower than the theoretical value. Adjusting the cross-sectional area of the simulated coil instead of the thickness to

be equal with the actual coil, resulted in a close match to the theoretical value with a relative error of 0.7%. Repeating the Maxwell simulation at 55 °C increased the resistance according to Equation (2.2). Simulating the solenoid in HFSS with one or two ports yielded the same results, both of which were slightly higher than the theoretical values.

Table 3.1: Inductance and resistance of the solenoid acquired by different methods.

Method	Inductance (μH)	Resistance at 22 °C (Ω)	Resistance at 55 °C (Ω)
Ideal	297	5.41	6.1
Wheeler formula	268	—	—
Maxwell (0.25mm)	263	3.56	—
Maxwell (adjusted)	264	5.45	6.15
HFSS (1&2 port)	275	6.34	—
Experimental	296	5.85	6.61

The experimental inductance was calculated to be 296 μH from the 0.654 MHz resonance peak with the 0.2 nF test capacitor (Figure 3.4). Although the higher frequency self-resonance peaks were visible in reflection mode (Figure 3.5), the tuned resonance peak was not observed and hence could not be used to determine the impedance of the coil. The fundamental self-resonant peak was at 5.73 MHz. The measured resistance was 5.85 Ω at 22 °C. The self-capacitance of the solenoid C_{self} was estimated to be 1.39 pF from the Medhurst formula $C_{\text{self}} = HD$ [37], where D is the diameter of the coil in cm and H is the dimensionless Medhurst factor (0.754 in this case) based on the length-to-diameter ratio of the coil.

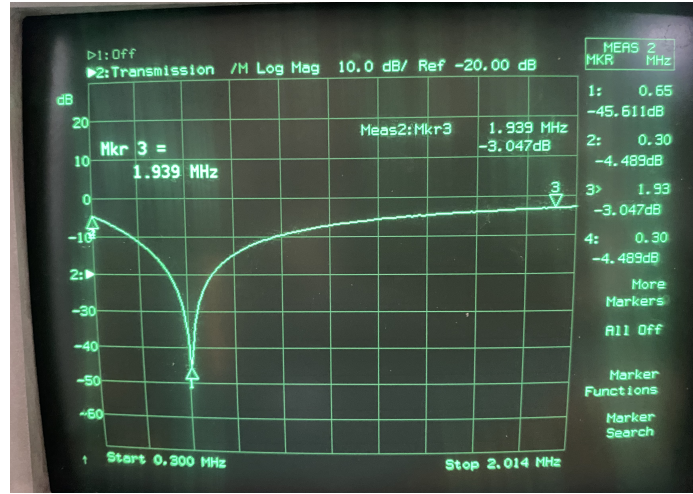


Figure 3.4: S_{21} (dB) plot for inductance measurement with a parallel 0.2 nF capacitor. Resonance frequency at 0.654 MHz.

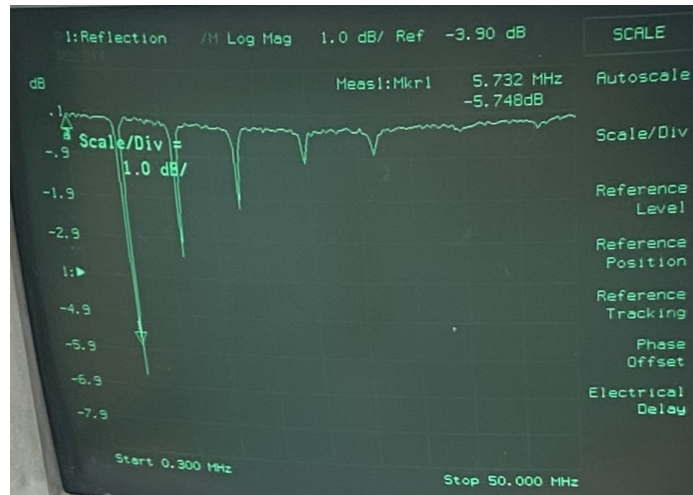


Figure 3.5: S_{11} (dB) plot for inductance measurement with a parallel 0.2 nF capacitor. Self resonance frequency peaks with a fundamental resonance at 5.73 MHz with no tuned resonance at 0.654 MHz.

3.2.2 Tuning and matching

The unloaded and loaded S_{11} (dB) and normalised impedance Smith charts are shown in Figures 3.6c and 3.6d and for comparison the simulated unloaded coil shown in Figures 3.6a and 3.6b. The unloaded and loaded traces nearly overlapped, resulting in the same 14.1 kHz resonance frequency after adding the L-section matching circuit to the coil (Figure 2.8). Their reflection coefficients at resonance were respectively -24.5 dB and -25.3 dB. Their frequency bandwidths were slightly different, 7.02 kHz

for the unloaded and 6.89 kHz for the loaded, while the respective quality factors were 2.01 and 2.04. The matched relative impedance for the unloaded coil was $1.11 - 0.066j$ and the loaded was $1.10 - 0.068j$. These values were comparable to the simulated matching characteristics which were 14.6 kHz resonance frequency, 7.3 kHz frequency bandwidth, Q_{unloaded} of 2 and a relative impedance of $1.02 + 0.04j$.

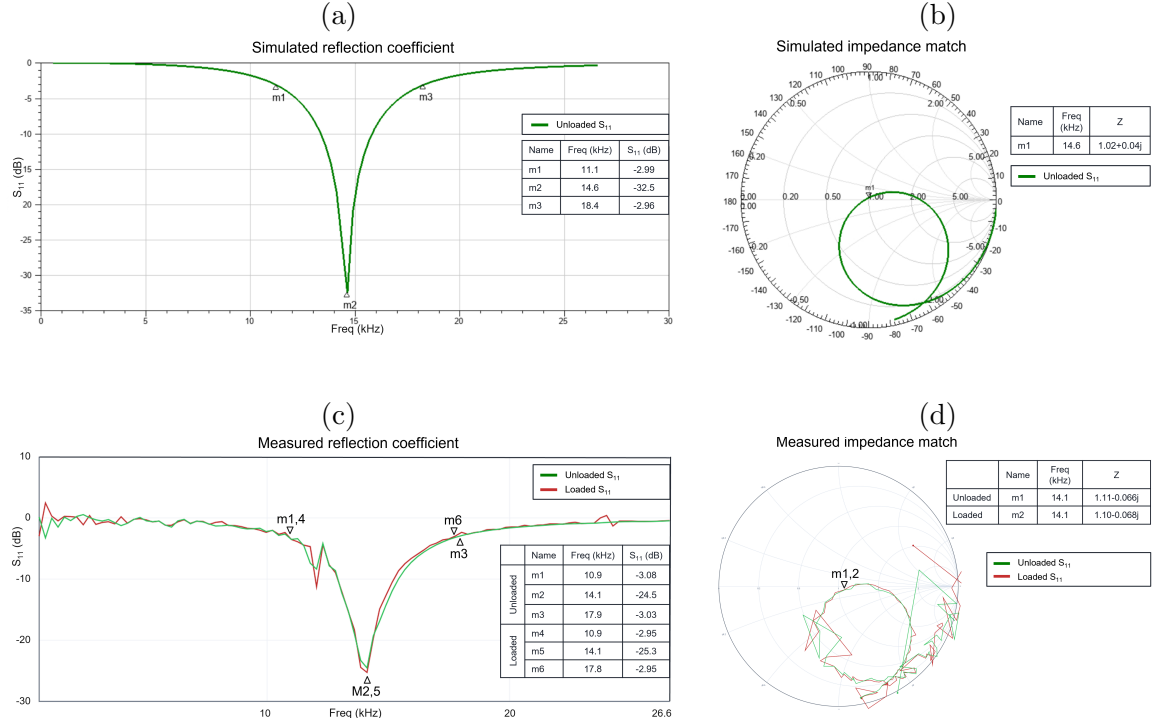


Figure 3.6: Tuning and matching outcome (a) of simulated Maxwell (adjusted) simulated S_{11} (dB) plot (b) and Smith chart. (c) Measured S_{11} (dB) plot (d) and Smith chart

As shown in Table 3.1, due to heating the resistance of the solenoid could increase from 5.85Ω to 6.61Ω , possibly detuning the coil. Between the different temperatures simulations, f_0 stayed 14.6 kHz. With the increase in resistance, the bandwidth broadened by 3.29%, and notably S_{11} (dB) at resonance frequency reduced by 20%.

Chapter 4

Discussion

4.1 Helmholtz coil

4.1.1 Heating

The main factor limiting the resonance frequency turned out to be the heating of the Helmholtz coil winding and the deflection temperature of the 3D printing material. Showcasing the significance of cryogenic cooling for high-field NMR and MRI scanners.

Temperatures above 55 °C could become problematic for cyanoacrylate glue which has a maximum service temperature of 82 °C. The heat generated would also diminish the polarisation of the sample.

4.1.2 Simulation

The Helmholtz coil produced a more homogeneous B_0 along its own axis rather than perpendicular to it. The coil homogeneity worsened further apart from its axis. The hotspots around the winding (Figure 3.2a) could constrain the effective detection volume. As the 10 mL Plastipak syringe diameter was 16 mm, these hotspots remained outside of its reach. The limiting factor for the effective detection volume was therefore B_0 homogeneity along the solenoid axis. $\Delta B_0/2$ was achieved within a 48.5 mm range, which corresponded to the 8 mL mark on the syringe, setting the effective detection volume. 8 mL is comparable to the middle sized probe of MQC-R, a commercially available NMR.

4.2 Transmit/receive solenoid

4.2.1 Heating

The PLA temperature would not be an issue for the T/R solenoid because to satisfy $B_1 \ll B_0$ [1, 6, 7] with a field-per-current of 4.05 mT/A_{peak}, the driving current would be at the lower end of the mA range. The simulations showed the resistance change due to heating between the tuning temperature and the maximum operating temperature was not large enough to drift the resonance frequency. The low Q and consequently the large bandwidth of the coil mitigated the consequences of frequency drift and made the fine tuning of the coil less important [19]. The used polypropylene film capacitors were relatively insensitive to temperature, but placing the tuning and matching capacitors outside the sample holder reduced the capacitors impact on B_0 and the capacitance drifting due to heat.

The winding of an earlier tested solenoid became tangled during testing, possibly due to shrinkage of the coil form at high temperatures. For this reason, the winding of the later coil was glued to the form across all turns. The 0.6 mm wall thickness of the solenoid coil form made it more prone to deformation. On the other hand, it was acceptable if the coil temperature exceeded 55 °C as there was no mechanical load on the structure. MQC used polytetrafluoroethylene (PTFE) for proton and acrylic coil for fluorine coil forms [23], Chowdhury et al. used fibreglass tubing [14, Supporting information].

4.2.2 Simulation

The Maxwell simulation of the solenoid with the adjusted winding thickness yielded a similar resistance and inductance values to the theoretical ones, with relative errors of -1% and -1.5% , respectively. HFSS simulations yielded a similarly accurate inductance with a 2.6% and a resistance of 17.2% relative error. HFSS simulates the skin effect and the proximity effect which would increase the resistance, but at 14.1 kHz both of these effects are negligible [32, 38]. Maxwell simulation with reflection mode could not reproduce the self-resonance peaks because of the implicitly simulated winding. However, it was able to show the absence of the tuned peak. The frequency range was too large and consequently, it was too computationally demanding to use HFSS to simulate self-resonance. The simulated magnetic field distribution was verified by both the theoretical value and the measurement at the top of the coil.

These results show that Maxwell is a faster and more accurate method of simulating a solenoid for our frequency range compared to HFSS, which verifies the recommendations to use Maxwell for lower frequency simulations, below 1 MHz, at which AC effects are negligible [38].

4.2.3 Resistance

The difference between the ideal resistance and the experimentally measured result can be explained by the total 78 cm of wire used to connect the coil winding to the multimeter. Accounting for this additional wire, the resistance changed from $5.41\ \Omega$ to $5.68\ \Omega$, bringing the relative error to 3%. It was unlikely that the coil had a larger number of turn than expected because the solenoid form could not fit more turns than 266.

4.2.4 Inductance test

The 10.4% discrepancy between the measured inductance and the one calculated from the Wheeler formula can be partially explained by the 5% tolerance of the multilayer ceramic capacitor used. The 1.39 pF self-capacitance of the solenoid would not significantly contribute to the measured inductance. The experimental inductance was measured to be nearly identical to the result of the ideal inductance formula. However, calculating the inductance with the Wheeler formula for this solenoid was justified as its dimensions fall within the optimal range of the Wheeler formula, where $l > 0.8r$ [35].

Performing the inductance test in reflection mode was not possible because the impedance of the coil was so poorly matched at the resonance frequency that it effectively acted as an open circuit and reflected all the signal, making the peak unobservable. The self-resonance peaks occurred because, the 15.4 m wire itself behaved as a quarter-wave resonant antenna creating standing waves. One end of the coil was grounded through the VNA, causing self-resonance peaks at frequencies given by $f = n \frac{c}{4l}$, where n is a positive integer, l is the wire length and c is the speed of light. The observed fundamental self-resonance peak of 5.73 MHz was higher than the expected 4.87 MHz possibly because inter-turn coupling increased the phase velocity along the coil axis above the speed of light [39]. A more straightforward explanation for the higher fundamental self-resonance frequency would be the shorter than calculated wire length. It would take the coil to have 2.33 m shorter wire or 40 turns less

than what is accounted for, which is highly unlikely because missing so many turns would have been noticeable on the T/R solenoid.

The higher self-resonance frequencies were visible because the skin effect and the proximity effect for the 0.25 mm diameter wire in the MHz range became significant and increased the effective resistance of the coil, thus bringing its impedance closer to reference impedance of 50 Ω . With even higher frequencies, the resistance increased, worsening the impedance match, and gradually flattening the higher-order resonance peaks. These self-resonance peaks are sufficiently far from the tuned frequency of 14.1 kHz to not interfere with it because no signal would be generated by the NMRduino at these frequencies.

4.2.5 Tuning and matching

The impedance of the solenoid was matched sufficiently well to the 50 Ω VNA impedance, as the reflection coefficients at f_0 for both the unloaded and loaded setups were below the commonly accepted threshold of -20 dB [40, 41]. Because the resonance frequency was only determined with a frequency resolution of 0.26 kHz, the measured reflection coefficient was slightly higher than the actual minimum value.

A Q_{loaded} of 2.04 is considered to be a very low value, as the typical range is 10-100. Q_{ratio} was approximately 1, indicating low sensitivity as the coil losses were dominated by its own resistance rather than the load. A higher Q_{ratio} could be achieved by reducing the resistance of the solenoid. The high signal from a hyperpolarised sample reduces the importance of the sensitivity in favour of a larger effective detection volume. One of the aims was to use the same sample at thermal polarisation as a known reference, in which case the coil sensitivity would be essential and Q_{ratio} would need to be increased beyond the recommended value of 2 [30].

4.3 Future work

For both the Helmholtz coil and the solenoid, using a larger diameter wire would be desirable to reduce the resistance and hence the heating, improve the signal-to-noise ratio or allow the increase of B_0 . The Helmholtz coil would need a larger spacing and hence a larger radius in order to accommodate a solenoid with a thicker winding. To keep the same B_0 field in the centre in case of a larger Helmholtz radius, the number of turns would need to be increased as well. As already implemented for the solenoid, using slits for the winding termination on the flanges would be preferred for the Helmholtz coil form as well instead of holes to make the winding

thickness more adjustable. Another way to reduce the limiting effects of heating would be to use 3D printing material with a higher heat deflection temperature, such as polyethylene terephthalate glycol (PETG) with 68 °C and the less common polypropylene (PP) glass fibre with 138 °C. PETG is a more accessible and practical option as exceeding 68 °C would start to become problematic for reasons such as sample heating, the maximum voltage output of the NMRduino of 10 V, and the maximum service temperature of the glue, 82 °C.

According to the current protocol a 50 mL falcon tube is used for the hyperpolarised tracer dissolution. Hence, it would be beneficial to design the sample holder to accommodate it. The larger diameter solenoid would reduce the tuning and matching capacitance to more manageable values, as L is directly proportional to R^2 . Such a coil setup would also increase the effective detection volume and simultaneously reduce the time between dissolution and polarimetry, increasing the measured signal by eliminating the step of transferring the tracer from the falcon tube to the 10 mL syringe.

The NMRduino output signal would need to be tested with an oscilloscope as it could not be done because the NMRduino user interface only became available after the end of this project [25]. There was room left for the interface box for a future T/R switch and pre-amplifier which will need to be matched to 50 Ω , proposed components for them can be found in Table A.2. For the pulse sequence the variable flip angle pulse sequence currently used in OCMR [15], could be adapted. Due to the high field-per-current of the T/R coil, there would be no need to extend the original 1 s long repetition time of the original pulse sequence. The polarimetry result of the final benchtop NMR will need to be compared with those obtained using the 3 T MRI for validation.

4.4 Conclusion

The aim of the project was to build a benchtop NMR for liquid-state ^{13}C polarimetry. Although an NMR console, a Helmholtz coil and a tuned and matched NMR probe were developed together with a methodology to simulate them, several further components would be needed before this system could be an alternative to the 3 T MRI scanner for polarimetry. These include a user interface to communicate with the NMR console, a T/R switch for the NMR probe, a pre-amplifier for the received signal, and electromagnetic shielding.

This thesis described, characterised and evaluated the developments made towards the benchtop NMR. A comparative analysis of HFSS and Maxwell demonstrated that Maxwell’s low-frequency eddy current solver provided a faster simulation and a slightly more accurate inductance value of the solenoid. The Maxwell simulation resulted in a relative error of -1.5% compared to the HFSS 2.6% , confirming that Maxwell was a more practical solver for frequencies below 1 MHz. However, both differed from the experimentally measured inductance. The Helmholtz coil generated a B_0 of 1.32 mT, corresponding to the targeted resonance frequency of 14.1 kHz, while remaining within the thermal limit of the 3D printed coil form. The constructed solenoid generated a field-per-current of $4.05 \text{ mT/A}_{\text{peak}}$ and was matched to 50Ω with an L-section circuit, yielding a reflection coefficient of -25.3 dB when loaded. The high wire resistance limited the system’s sensitivity, resulting in a low Q_{loaded} of 2.04.

Overall, the proposed benchtop NMR was promising for relative liquid-state polarimetry, but future optimisations and testing are needed before it could relieve the 3 T MRI scanner for quantitative polarimetry.

Statement of AI use

I acknowledge the use of ChatGPT 5.6Sol (OpenAI, [Link to website](#)) and Overleaf AI assistant (Digital Science UK Limited, [Link to website](#)) and Gemini 3.6 Flash (Google, [Link to website](#)) for creating preliminary plot of Figure 3.1 (Helmholtz coil heating) in Python to match Ansys rectangular plot style, proofreading thesis draft, formatting in LaTeX, and searching for relevant publications.

Prompt examples:

- Write a Python code to plot this data in the style of the one on the attached image.
- List any spelling errors, grammatical errors, undefined acronyms or inconsistent symbol use with recommended corrections in this text.
- How do I make an URL appear as hyperlink in LaTeX?
- Could you recommend a paper with a use case of Maxwell and HFSS?

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Appendix A

Component list

Table A.1: Component list.

Component	Unit	Source
Integrated chips:		
DRV8838 (Pololu, USA; 2990)	2	Pololu Link
DRV8835 (Pololu, USA; 2135)	2	Pololu Link
MCP4822-E/SN (Microchip Tech., USA)	1	RS Online Link
L272D (STMicroelectronics, Switzerland)	1	RS Online Link
LTC1859CG#PBF (Analog Devices, USA)	1	Analog Devices Link
Connectors:		
Pin socket (Samtec, USA; SSA-115-S-T)	78 contacts	RS Online Link
Pin header (Wurth Elektronik; 61301611121)	78 contacts	RS Online Link
SMA F. PCB mount (RS, UK; 546-3276)	7	RS Online Link
BNC F. (Amphenol RF; B6251C1-NT3G-50)	3	RS Online Link
SMA F. panel mount (RS, UK; 921-0215)	5	RS Online Link
Solder tag (RS, UK; 186-7826)	5	RS Online Link
Capacitors:		
0.1 μ F ceramic, 0603 (Vishay, USA)	4	RS Online Link
1 μ F ceramic, 1206 (KEMET, USA)	1	RS Online Link
10 μ F ceramic, 1206 (Murata, Japan)	4	RS Online Link
0.1 to 1 nF ceramic (PPI, UK; DKD1111P07)	1	Passive Pluse Link
1 to 1000 nF film (AUKENIEN, China)	1	—
Cables:		
M-M BNC 50 Ω RG58 (RS, UK; 284-3792)	2	RS Online Link
M-M SMA 50 Ω RG316 (RS, UK; 498-7585)	5	RS Online Link

Table A.2: Component list continuation.

Component	Unit	Source
Other:		
Printed circuit board	1	GitHub Link
Teensy 4.1 (SparkFun Electronics, USA)	1	SparkFun Link
Brass threaded insert M3 (RS, UK; 204-0616)	4	RS Online Link
Nylon machine screw (Würth Elektronik; 97790503111)	4	RS Online Link
0.25 mm copper wire (RS, UK; 357716)	41 m	RS Online Link
3D printed components (STL files and models)	-	GitHub link
Proposed components:		
MD0100N8-G, SOT (Microchip Tech, USA)	1	DigiKey Link
PA0084, SOT-89 to DIP (Chip Quik Inc., Canada)	1	DigiKey Link
OPA134PA (Texas Instruments, USA)	1	RS Link
O-ring (RS, UK; 196-5377)	4	RS Link

Appendix B

Permanent magnet simulation results

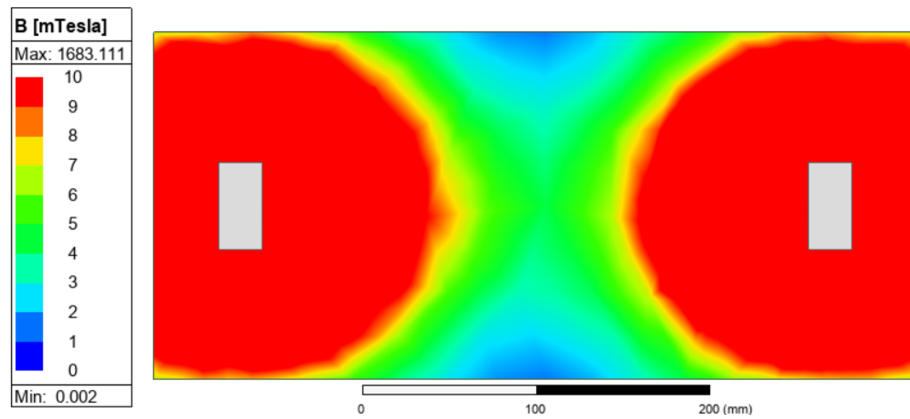


Figure B.1: Permanent magnet field homogeneity (heat map).

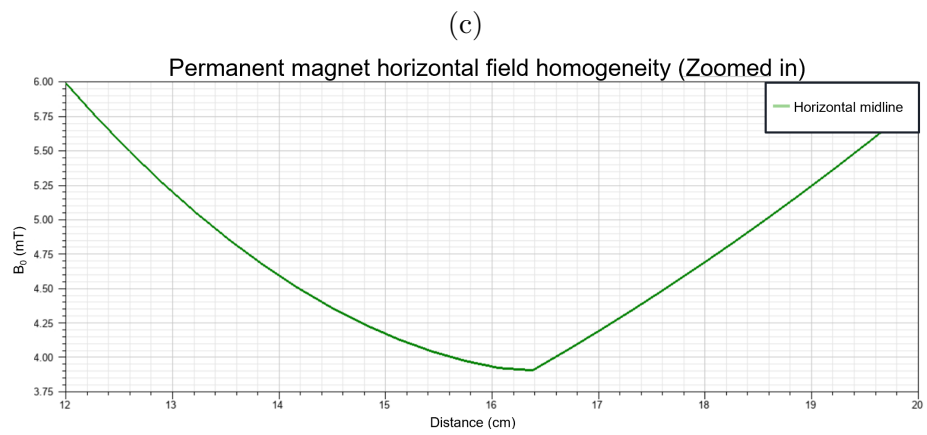
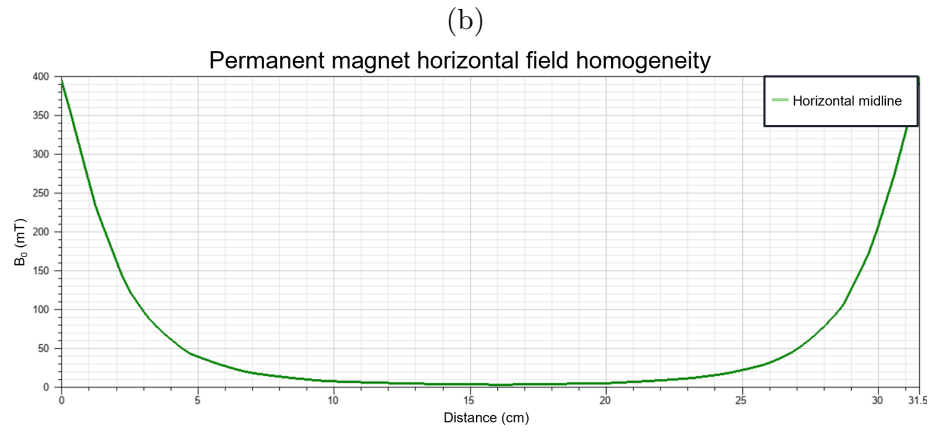
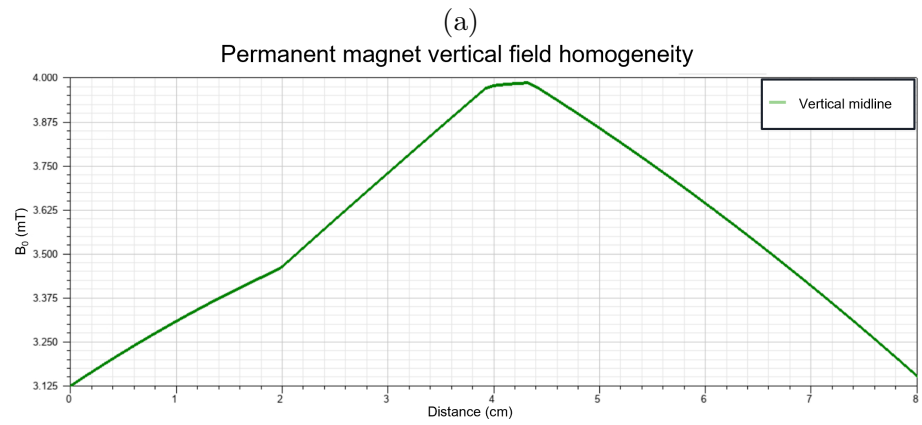


Figure B.2: Permanent magnet field homogeneity along the (a) vertical midline (b) horizontal midline and (c) horizontal midline zoomed in on the centre.